CBE 142 Cheat Sheet

BSTR

$$r_j \int V = \frac{dN_j}{dt}$$

$N = C \cdot V$ ($V \neq$ const)

$$\therefore \frac{dN_j}{dt} = V\frac{dC_j}{dt} + C_j\frac{dV}{dt}$$

V = const: $r_j = \frac{dC_j}{dt}$

Design eqs:

* $r_A = -C_A \frac{dX}{dt}$
* $\frac{t}{C_{A0}} = \int_0^{X_A} \frac{dX_A}{(-r_A)}$
* $dt = \frac{-N_{A0}\,dX}{V \cdot r_A}$

Mole Balance: In − Out + generation = Acc.

$$F_{j0} - F_j + \int_V r_j\,dV = \frac{dN_j}{dt}$$

Ergun eqn: $\frac{dP}{dW} = \frac{-\alpha P_0^2}{2P}(1+\varepsilon X)\frac{T}{T_0}$

Packed bed: $\frac{dF_A}{dW} = -r_A' \Rightarrow$ if P = const $\Rightarrow \int_{F_A}^{F_{A0}} \frac{dF_A}{-r_A'} = W$

Membrane: $R_B = k_B' a_v (C_B - 0) \simeq k_B C_B$

$$\frac{dF_B}{dV} = r_B - R_B$$

$W = \rho_b V$

Mass balance:

$$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out}$$

$$\frac{d(\rho V)}{dt} = \rho_0 v_0 - \rho_0 v$$

$\dot{m}_i = (F_i)(MW)_i$

$\dot{m}_i = (C_i \cdot v)(MW)_i$

PFR

$$r_j = \frac{F_j - F_{j0}}{\Delta V}; \quad r_j = \frac{dF_j}{dV}$$

V = const; $r_j = v_0 \frac{dC_j}{dV}$ or $d\tau = \frac{dC_j}{r_j}$

$v \neq$ const; $dV = d\tau \cdot v(x,y)$

$d\tau = dV / v_0$

$$\tau \frac{V}{F_{A0}} = \frac{\tau}{C_{A0}} = \int_0^{X_A} \frac{dX_A}{-r_A}$$

* $r_A = -F_{A0}\frac{dX}{dV}$
* $V = \int_0^X \frac{F_{A0}}{-r_A} dX_A$

$\frac{dF_A}{d(\rho V)} = -r_A' \Rightarrow \frac{dF_A}{dV} = \rho r_A'$

CSTR

$$r_j \cdot \int V = F_j - F_{j0} \quad \text{(steady state)}$$

$v_0 = $ cont $V = \frac{v_0(C_{j0} - C_j)}{-r_j}$

First order SS: $X = \frac{k\tau}{1+k\tau} = \frac{Da}{1+Da}$

* $\tau = \frac{C_{A0} \cdot X_A}{-r_A}$
* $V = \frac{F_{A0} \cdot X_A}{-r_A}$

(series) $V = \frac{F_{A0}(X_f - X_i)}{-r_A}$

$\tau = \frac{V}{v_0}$; $F = C \cdot v$; $N = C \cdot V$

$N = F \cdot t$

$$r_{net} = \frac{r_1 - r_{-1}}{\sigma_1} = \frac{r_2}{\sigma_2} = \frac{r_3 - r_{-3}}{\sigma_3} = \frac{r_4}{\sigma_4}$$

$Da = \tau k$

$Da = \frac{-r_{A0}V}{F_{A0}} = \frac{k_1 C_{A0} V}{v_0 C_{A0}}$ (Reactn rate of A) / (Convection rate of A)

$Da < 0.1 \Rightarrow X < 0.1$

$Da > 10 \Rightarrow X > 0.9$

By stoich: (Can use both)

$aA + bB \rightarrow cC + dD$

$$\frac{-r_A}{a} = \frac{-r_B}{b} = \frac{r_C}{c} = \frac{r_D}{d}$$

$aA + bB \rightarrow cC + dD$ [Batch]

Species	Initial	Change	Final
A	N_{A0}	$-N_{A0}X_A$	$N_{A0}(1-X_A)$
B	N_{B0}	$-b/a\, N_{A0}X_A$	$N_{B0} - b/a\, N_{A0}X_A$
C	N_{C0}	$c/a\, N_{A0}X_A$	$N_{C0} + c/a\, N_{A0}X_A$
D	N_{D0}	$d/a\, N_{A0}X_A$	$N_{D0} + d/a\, N_{A0}X_A$

$$\delta = \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - \frac{a}{a}\right)$$

$$\varepsilon = y_A \cdot \delta$$

$$\Theta_B = \frac{F_{B0}}{F_{A0}} \quad ; \quad y_A = \frac{F_{A0}}{F_{T0}}$$

$aA + bB \rightarrow cC + dD$ [Flow & Condensation]

Species	Initial	Change	Final (b·c)	Final (a·c)
A	F_{A0}	$-F_{A0}X_A$	$F_{A0}(1-X_A)$	$F_{A0}(1-X_A)$
B	F_{B0}	$-b/a\, F_{A0}X_A$	$F_{B0} - b/a\, F_{A0}X_A$	$F_{B0} - b/a\, F_{A0}X_A$
C	F_{C0}	$c/a\, F_{A0}X_A$	$F_{C0} + c/a\, F_{A0}X_A$	$F_{C0} + c/a\, F_{A0}X_A$
D	F_{D0}	$d/a\, F_{A0}X_A$	$F_{D0} + d/a\, F_{A0}X_A$	$y_D^* F_T$

$$v = v_0\left(\frac{F_T}{F_{T0}}\right)\left(\frac{T}{T_0}\right)\left(\frac{P_0}{P}\right) \qquad \frac{F_T}{F_{T0}} = (1 + \varepsilon X_A) \qquad F_T = F_{T0} + F_{A0}\delta_2$$

$$V = V_0\left(\frac{N_T}{N_{T0}}\right)\left(\frac{T}{T_0}\right)\left(\frac{P_0}{P}\right) \qquad F_A = F_{A0}(1-X_A)$$

$$N_A = N_{A0}(1-X_A)$$

$$C_A = C_{T0}\left(\frac{F_A}{F_T}\right)\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right) \rightarrow \text{PFR, PBR}$$

$$k(T) = k(T_{ref})\exp\left(\frac{-E_a}{R}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right) \rightarrow \text{Finding } k \text{ at a diff } T \text{ from reference } k.$$

$$k = A\exp\left(\frac{-E_a}{RT}\right) \rightarrow \text{Arrhenius equation}$$ [First order elementary reactions obey this]

Ex: $P + Q \rightarrow 2R + S$ [Isothermal BSTR]

Higher X_A : Const V BSTR → More conc. of reactant.

$P + Q \rightleftharpoons 2R + S$

$$K = \frac{[N_R^2 N_S]}{[N_P N_Q]}\frac{1}{V} = \text{const.}$$

If V↑, $N_R^2 N_S$ ↑ : V is helping. P is const.

$2A + B \rightleftharpoons C + D$

$$K = \frac{N_C N_D}{N_A^2 N_B} V = \text{const}$$

V↑, $N_C N_D$ ↑

V is helping, such P

Energy Balance

Equilibrium Conversion:

for a reversible reaction, X_e is the maximum conversion that can be achieved at the reaction temp.

eg: $N_2O_4 \rightleftharpoons 2NO_2$

$$K_c = \frac{[NO_2]^2}{[N_2O_4]} = \frac{C_B^2}{C_A}$$

From stoich table, $C_A = C_{A0}(1-X_A)$

$C_B = 2C_{A0}X_A$

$$\therefore K_c = \frac{4C_{A0}^2 X_A^2}{C_{A0}(1-X_A)}$$

$$K_c = \frac{4C_{A0}X_A^2}{(1-X_A)}$$

This is the equilibrium conversion

$$\therefore \boxed{X_e = X_A = \sqrt{\frac{K_c(1-X_A)}{4C_{A0}}}}$$

+ For a first order reversible reaction: $A \rightleftharpoons B$

$$K_c = \frac{C_B}{C_A} = \frac{C_{A0}(X_A)}{C_{A0}(1-X_A)}$$

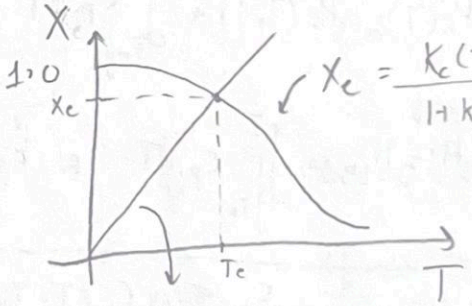
$$K_c(1-X_A) = X_A$$

$$K_c - K_cX_A = X_A$$

$$\boxed{X_e = X_A = \frac{K_c}{1+K_c}}$$

The equilibrium conversion is dependent on T of reactor

↳ For an exothermic reaction, as T ↑, the K_c ↓ and therefore X_e ↓



$$X_e = \frac{K_c(T)}{1+K_c(T)}$$

When solving questions that are non-isothermal, it is important to understant that K_c, X_e and X_{eb} are all temp. dependent.

X_e → mole balance

X_{EB} → energy balance

} The intersection of these two provids the adiabatic equilibrium Te and Xe (Reactor conditions)

$$X_{EB} = \frac{\Sigma\Theta C_p(T-T_0)}{-\Delta H_R}$$ (for adiabatic reactor)

$$= \frac{C_{P_A}(T-T_0)}{-\Delta H_{R_A}}$$ (Constan C_p)

$$X_{EB} = \frac{C_{P_A}(T-T_0)}{-\Delta H_{R_A}}$$ } Linear equation

Energy balance equations:

$$\text{Energy Acc} = \dot{Q} - \dot{W} + \text{energy in} - \text{energy out}$$

$$\frac{d\hat{E}_{sys}}{dt} = \dot{Q} - \dot{W} + \sum_{i=1}^{n} E_i F_i \Big|_{in} - \sum_{i=1}^{n} E_i F_i \Big|_{out}$$

where, $E_i = U_i + \frac{u_i^2}{2} + g z_i$

$\dot{W}$ → Flow work: $F_i \cdot P\bar{V}_i$; Shaft work: $\dot{W}_s$

If we let $E = U$ & take F_i common, $F_i(U_i + P\bar{V})$ ($U_i + P\bar{V} = H$) we get the energy balance as:

$$\frac{d\hat{E}_{sys}}{dt} = \dot{Q} - \dot{W}_s + \sum_{i=1}^{n} F_i H_i \Big|_{in} - \sum_{i=1}^{n} F_i H_i \Big|_{out}$$

expressing $F_A = F_{A0}(1 - X_A)$; $F_B = F_{A0}\left(\Theta_B - \frac{b}{a}X\right)$ & Adding up enthalpies to give ΔH_R

$$\sum_{i=1}^{n} F_{i0} H_{i0} - \sum_{i=1}^{n} F_i H_i = F_{A0} \sum_{i=1}^{n} \Theta_i (H_{i0} - H_i) - \Delta H_{R_x}(T) F_{A0} X$$

$$\therefore \boxed{\dot{Q} - \dot{W}_s + F_{A0} \sum \Theta_i (H_{i0} - H_i) - \Delta H_{R_x}(T) F_{A0} X = 0}$$

→ S.S
→ For known H's & ΔH_R
→ Can be used for phase change

If no phase change:

$$H_i(T) = H_i^{\circ}(T_{ref}) + \int_{T_{ref}}^{T} C_{P_i} dT \quad \text{(reference temp. \& enthalpies cancel)}$$

$$H_i - H_{i0} = \int_{T_{i0}}^{T} C_{P_i} dT \approx C_{P_i}(T - T_{i0}) \quad \rightarrow \text{Assuming } C_p \neq T \text{ dependent}$$

★ $$\therefore \boxed{\dot{Q} - \dot{W}_s - F_{A0} \sum \Theta C_{P_i}(T - T_{i0}) - \Delta H_R(T) F_{A0} X = 0}$$ ★

→ S.S
→ When cp's are know but H is not known
→ No phase change.
→ C_p is constant w/ T

where, $\Delta H_R(T) = \Delta H_R^{\circ}(T_{ref}) + \Delta C_p(T - T_{ref})$

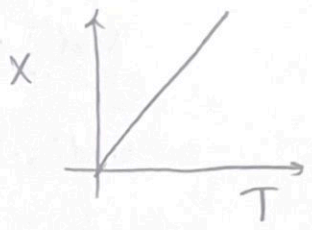
Adiabatic energy balance

$\dot{W}_s = 0$, $\dot{Q} = 0$ & $T_{i0} = T_0$, we can rearrange the previous OP OG equation

to give:

$$X = \frac{\sum \theta_i C_{P_i}(T - T_0)}{-\Delta H_R(T)} = \frac{\sum \theta_i C_{P_i}(T - T_0)}{-[\Delta H_R^\circ(T_{ref}) + \underbrace{\Delta C_p(T - T_{ref})}_{\text{negligible}}]} \approx \boxed{\frac{\sum \theta_i C_{P_i}(T - T_0)}{-\Delta H_R^\circ(T_{ref})}}$$

Linear plot.



(Valid for BSTR, CSTR, PFR & PBR)

↓ solving for T

$$T = \frac{X[-\Delta H_R^\circ(T_{ref})] + \sum \theta_i C_{P_i} T_0 + X \Delta C_p T_R}{\sum \theta_i C_{P_i} + X \Delta C_p}$$

This can now be used to solve for r_A

eg: [PFR] $A \underset{k_2}{\overset{k_1}{\rightleftharpoons}} B$

$$-r_A = +k_1 C_A - k_2 C_B = k_1\left[C_A - \frac{k_2 C_B}{k_1}\right] = k_1\left[C_A - \frac{C_B}{K_c}\right]$$

$$C_A = C_{A0}(1 - X)\frac{T_0}{T}$$

$$C_B = C_{A0} X_A \frac{T_0}{T} \quad \therefore -r_A = k C_{A0}\left[(1 - X) - \frac{X}{K_c}\right]\frac{T_0}{T}$$

; $k_1(T) = k \exp\left[\frac{E}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right]$

; $K_c(T) = K \exp\left[\frac{-\Delta H_R^\circ}{R}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right]$

; $T = \frac{X[-\Delta H_R^\circ(T_{ref})] + \sum \theta_i C_{P_i} T_0 + X \Delta C_p T_R}{\sum \theta_i C_{P_i} + X \Delta C_p}$

∴ We have two knowns (X, V)

eqn: 1) $\frac{dX}{dV} = \frac{-r_A}{F_{A0}}$ } ODE solver!

2) $k(T)$

3) $K_c(T)$

4) $T(X)$

If $\Delta C_p = 0 = C_{P_B} - C_{P_A} = 0$

$$= \frac{X[-\Delta H_R^\circ(T_{ref})]}{\sum \theta_i C_{P_i}} + T_0$$

Wall cooled / Heat exchanges w/ reactors

With the addition of extra heat, the energy balance will now include a Q term that is due to heating.

$$\Delta\dot{Q} + \Sigma F_i H_i\big|_V - \Sigma F_i H_i\big|_{V+\Delta V} = 0 \quad ; \quad \Delta\dot{Q} = U\Delta A(T_a - T)$$

(heat trans → U; heat exchange area → ΔA; ambient temp. → T_a)

Therefore the differential equation that we get:

$$U_a(T_a - T) - \frac{d\Sigma(F_i H_i)}{dV} = 0$$

$$\Delta\dot{Q} = U_a \Delta V (T_a - T)$$

(U_a: heat exchange area per unit volume)

$$a = \frac{A}{V} = \frac{\pi DL}{\frac{\pi D^2 L}{4}} = \frac{4}{D}$$

$$\Rightarrow U_a(T_a - T) - \Sigma \frac{dF_i}{dV} H_i - \Sigma F_i \frac{dH_i}{dV} = 0$$

PFR (Steady state)

$$\Delta H = c_p \Delta T \; ; \; \frac{dH}{dV} = c_p \frac{dT}{dV}$$

$$\therefore \boxed{\frac{dT}{dV} = \frac{r_A \Delta H_R - U_a(T - T_a)}{\Sigma F_i C_{P_i}}} = \boxed{\frac{r_A \Delta H_R - U_a(T - T_a)}{F_{A_0}(\Sigma \Theta_i C_{P_i} + \Delta C_p X)}}$$

$Q_g = r_A \Delta H_R$ (Heat generated) $\qquad Q_R = U_a(T - T_a)$ (Heat removed)

ρ_b → bulk density

Similarly, for a PBR: $\frac{dT}{dW} = \frac{r_A \Delta H_R - \frac{U_a(T - T_a)}{\rho_b}}{\Sigma F_i C_{P_i}}$

↓

Cocurrent

EB on heat transfer fluid:

$$\frac{dT_a}{dV} = \frac{U_a(T - T_a)}{\dot{m}_c C_{P_c}}$$

c → coolant/heating medium

Countercurrent

$$\frac{dT_a}{dV} = \frac{U_a(T_a - T)}{\dot{m}_c C_{P_c}}$$

Not overall energy balances ↓ Just on coolant

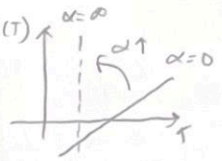
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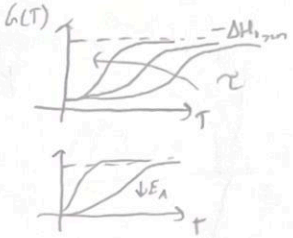
CBE 142

1) MSS: $G_i(T) = -\Delta H^\circ_{rxn} X_A$

$R(T) = C_{P_0}(T_a - T)$ ← From Newton's Law of cooling

$\alpha = \frac{\text{heat exchange}}{\text{heat going to fluid}} = \frac{UA}{F_{A0}\bar{C}_{P_0}} \cdot \frac{T - T_a}{T - T_0}$





↑τ, more opp to release heat during the reaction

2) Unsteady engy balas:

$$UA(T_a - T) - \underbrace{\sum_i F_{i0}(H_i - H_{i0})}_{F_{A0}\sum \Theta_i \tilde{C}_{p_i}(T - T)} - \underbrace{\sum_i \frac{\nu_i}{\nu_A} \tilde{H}_i r_A V}_{-\Delta H_{rxn}} = V \sum \tilde{C}_i C_i \frac{dT}{dt}$$

} For CSTR OP OG

Energy balance coupled w/ mass balance

Average C_p → $\hat{\Delta C_p} = \sum_i \frac{\nu_i}{\nu_A} \tilde{C}_{p_i}$

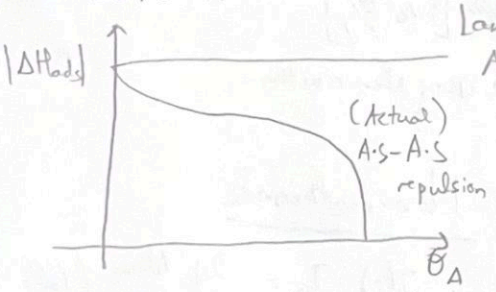
3) Catalysis & Selectivity:

(QE) $r_{ads} = r_{desorption}$; $C_{Tot} = C_{empty} + C_{occupied} = C_{tot} A_{A\cdot S}$; Θ_A = fractional coverage = fraction of total no. of sites

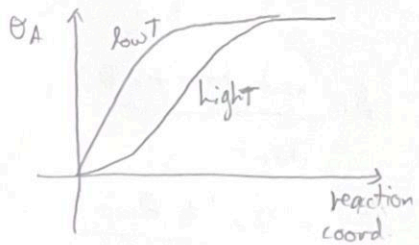
$k_A P_A C_S = k_{-A} C_{A\cdot S}$

$$\Theta_A = \frac{C_{A\cdot S}}{C_{Tot}} = \frac{K_A P_A}{1 + K_A P_A}$$

Langmuir-Isotherm:



Langmuir Adsorption } Assumes
1) when stuff adsorbs it does not move around
2) Adsorption sites nearby are not affected by the adsorption (in actual there is slight repulsion)



High energy → not easy to attach to the surface due to high internal energy

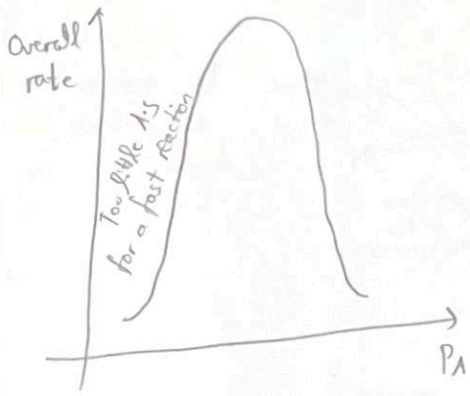
acts as a conc. gradient to pull the rate to equil

$$r = \frac{(\text{forward kinetic rate term}) - \left(1 - \begin{matrix}\text{normalized distance from} \\ \text{equil. for the reverse term}\end{matrix}\right)}{(\text{adsorption term})^n}$$

$$= \frac{k_S K_A K_B P_A P_B C_{tot}\left(1 - \frac{P_C}{P_A P_B}\frac{K_{rxn}}{}\right)}{(1 + K_A P_A + K_B P_B + K_C P_C)^2}, \quad K_{rxn} = \frac{k_S}{k_{-S}}$$

(MASI) → Most Abundant surface intermediate
↳ Lets you simplify C_{Tot} to the where the sites are most abundant that will be only included in C_{Tot}

+ Mass is on site balance (Shows up in the denominator of η)

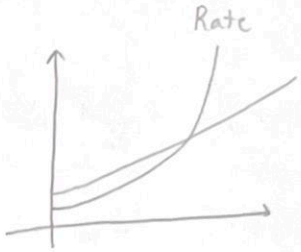


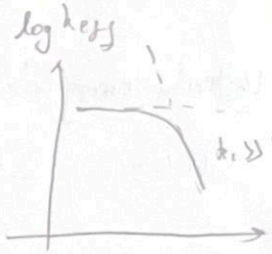
+ $k_{app.}$, $E_{app.}$ $\quad E_{app} = E_S + \Delta H_A^\circ + \Delta H_B^\circ$

$$r = \underbrace{k_S \Gamma_A \Gamma_B}_{k_{app.}} P_A^{\alpha} P_B^{\beta} C_{tot}$$

Film Theory

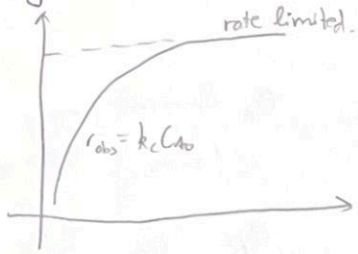
$$\frac{1}{k_{eff}} = \frac{1}{k_c} + \frac{1}{k_r} \quad ; \quad -r_A'' = \frac{k_r k_c C_{Ab}}{k_c + k_r} = k_{eff} C_{Ab}$$





momentum diff / Convective mass diff

→ Hydrodynamics $\quad k_c = f(Re, Sc)$



When $\Phi_n \sim 1$ (have to solve ODE)

↳ We non dimensionalize to get

$$\phi = \frac{C_A}{C_{As}} = \frac{1}{\lambda} \frac{\sinh}{\sinh}$$

η = Productivity of the catalyst. (Only depends on the rate) } also only internal

We want this close to 1 but not too close because we are mass transfer limited

Internal diffusion (Porous Catalyst)

+ Shell balance

$$\Phi_n^2 = \frac{R k_n C_{As}^n A_i}{D_{eff} \Xi_i \left[(C_{As} - 0)/R\right]}$$

(neglects external diffusion!)

$\Phi^2 << 1$ → very slow reaction

Non-isothermal

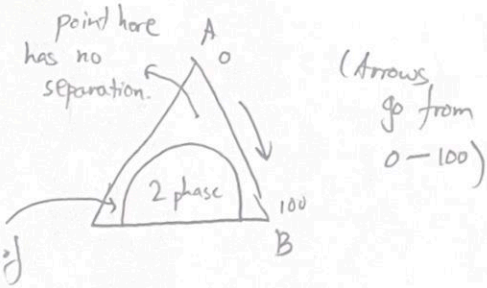
$$T(r) - T_s = \frac{D_{eff} \Delta H_{rxn}}{k_{eff}} (C_A(r) - C_{As})$$

$$\beta = \frac{\text{heat from reaction}}{\text{heat}} = -\frac{D_{eff} C_{As} \Delta H_{rxn}}{k_{eff}}$$

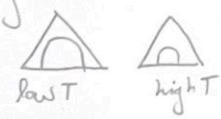
Chapter 15 CBE 150B

* Hunter-Nash

Nernst Law → driving force for LLE (eq. between two liquids due to conc.) } eq of solute due to different solubilities in phases.



Solubility $\propto T$



Component MB: $\frac{F}{S} = \frac{(w_c^S - w_c^M)}{w_c^M - w_c^F} = \frac{\overline{MS}}{\overline{FM}}$

When we don't know $E, \Delta R_N$?

↳ 1) $M = F + S = E_1 + R_N$

2) $M(x_i)_M = (y_i)_1 E + (x_i)_N R_N = (x_i)_F F + (y_i)_S S$

M is known (As F & S are know)

$(x_i)_F$ & $(y_i)_S$ are known

→ Find $(x_i)_M$

Now you can get (y_i), & $(x_i)_N$

Operating lines always start @ pivot point P

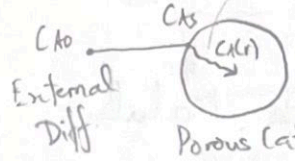
CBE 110 : Midterm 2 Review

CBE 142: Final Review

Chapter 15

$-D_{AB}\frac{d^2C_A}{dr^2} + r_A = 0$ [Mole balance including diffusion and reaction]

Internal Diffusion



Ext. Diffusion coefficient (in bulk fluid & BL)

Effective Diffusion: The catalyst itself has pores that twist / turn (tortuous). There are voids and solids of varying conc and diffusion does not always occur normal to the direction of flux.

↳ Therefore we define an effective Diffusion

$$D_e = \frac{D_{AB}\phi_p\sigma_c}{\tau}$$

- D_{AB}: Bulk diffusivity
- ϕ_p: porosity = Volume of empty (voids) / Total volume
- σ_c: constriction factor (Accounts for variation in normal cross-section area)
- τ: tortuosity = Actual distance travelled / Shortest distance possible

$r_A \Rightarrow \left[\frac{mol}{dm^3\,s}\right]$, $r_A' \Rightarrow \left[\frac{mol}{g_{catalyst}\,s}\right]$; $r_A'' \Rightarrow \left[\frac{mol}{m^2\,s}\right]$ $\therefore -r_A = \rho_c(-r_A')$

$-r_A = \rho_c S_a(-r_A'')$

$\frac{g_{catalyst}}{Volume}$ $\frac{m^2}{g_{catalyst}}$

↳ This is the same for rate constant units (k_1, k_1', k_1'')

Shell balance on porous catalyst:



(In @ r) − (Out @ r+Δr) + (gen within Δr) = 0

Flux Area − Flux·Area + Rate/mass · ρ_c · V_{shell} · Δr = 0

$W_A \times 4\pi r^2\big|_r - W_A \times 4\pi r^2\big|_{r+\Delta r} + (r' \cdot \rho_c \cdot 4\pi r_m^2 \Delta r) = 0$

↳ mean radius between r & r+Δr

$\div 4\pi\Delta r$

$\Delta r \to 0 \Rightarrow \frac{d(W_A r^2)}{dr} - r_A'\rho_c r^2 = 0$

Chain rule + Dividing by $r^2 D_e$

$W_A = -D_e\frac{dC_A}{dr}$ → moles of A / open-pore volume

$-r_A' = k_n''S_a\rho_c C_A^n$; k_n

$$\therefore \boxed{\frac{d^2C_A}{dr^2} + \frac{2}{r}\left(\frac{dC_A}{dr}\right) - \frac{k_n}{D_e}C_A^n = 0}$$

BC's: at $r = R$, $C_A = C_{As}$

$r = 0$, $\frac{dC_A}{dr} = 0$

$\psi = \frac{C_A}{C_A}$ $\tau = \frac{r}{R}$

Now if we non-dimensionalize, $\psi = \frac{C_A}{C_{As}}$; $\lambda = \frac{r}{R}$; $W_{A,r} = -D_e \frac{dC_A}{dr}$

Doing this gives us, $\boxed{\frac{d^2\psi}{d\lambda^2} + \frac{2}{\lambda}\frac{d\psi}{d\lambda} - \underbrace{\frac{k_n R^2 C_{As}^{n-1}}{D_e}}_{\text{This expression pops out}} \psi^n = 0}$

$k_n = k_n'' S_a \rho_c$

$\frac{-D_e d(\psi C_{As})}{d(R\lambda)} = -\frac{D_e C_{As}}{R}\frac{d\psi}{d\lambda}$

$\Rightarrow \phi_n^2$ Thiele Modulus

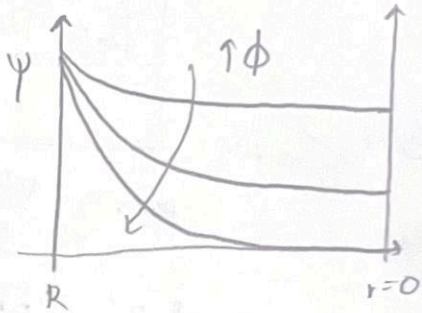
n → reaction order

$\therefore \phi_n \gg 1$, The reaction rate is very fast (Internal Diffusion limiting)

$\frac{\text{Surface rxn rate}}{\text{Internal diffusion rate}}$

We can solve the above Diff eq to get a solution for first order reaction:

$$\psi = \frac{A_1}{\lambda}\cosh\phi_1\lambda + \frac{B_1}{\lambda}\sinh\phi_1\lambda \Rightarrow \psi = \frac{1}{\lambda}\frac{(\sinh(\phi_1\lambda))}{(\sinh(\phi_1))}$$



As $\phi \uparrow$, the rxn rate increases and therefore A reacts faster than it diffuses going to zero @ the center

Effectiveness Factor

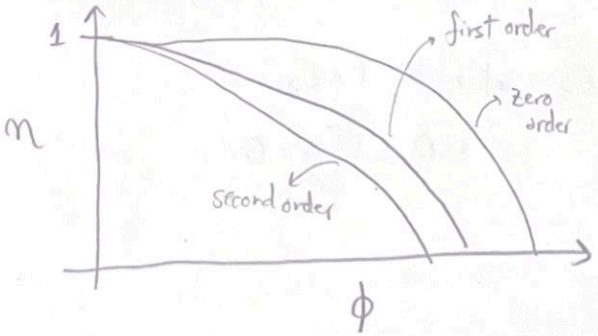
The reaction rate depends on conc. The conc depends on r along the catalyst. The rate also changes and the rate @ surface is not the same as rate in the middle } η (effectiveness factor)

$= \frac{\text{Actual rate}}{\text{Max rate possible if everything is @ } C_{As}, T_s}$

$\eta = \frac{M_A}{M_{As}}$; reaction rate $\left(\frac{\text{mol}}{\text{s}}\right)$; reaction rate

$M_A = -\frac{D_e C_{As}}{R}\left.\frac{d\psi}{d\lambda}\right|_{\lambda=1}(4\pi R^2)$; $M_{As} = k_1 C_{As}\left(\frac{4}{3}\pi R^3\right)$

We have to solve for η for every order by solving the diff eq.



As ϕ decreases, $\eta \to 1$ (surface rxn limited)

$\phi \propto R, \frac{1}{D_e}, k$

$\eta \propto \frac{1}{\phi}$

We can express rate in terms of Φ:

$$-r_A = \left(\frac{\text{Actual rate}}{\text{rate @ } C_{As}}\right) \times (\text{Rate @ } C_{As}) \quad , \quad -r_A = \eta(kC_{As})$$

$$\boxed{-r_A = \eta(-r_{As})}$$

effectiveness factor is a measure of surface accessibility

As $\Phi = \frac{\text{surface rxn rate}}{\text{internal diffusion MT}}$ ~~limited~~

$\Phi \gg 1$ means that the system is internal diffusion limited (surface reaction happening super fast)

↓

In this limiting case, η can be approximated as:

$$\boxed{\eta = \left(\frac{2}{n+1}\right)^{1/2} \frac{3}{\Phi}} \qquad \Phi \gg 1.$$

For a first order (Φ_1)

$$\eta = \frac{3}{\Phi_1^2}(\Phi_1 \coth(\Phi_1 - 1)$$

Chapter 14

+ For solid catalysts (non-porous) in PBR we can just use the PBR balance:

$$\frac{dF_A}{dW} = -r_A$$

(reaction happening only on the surface)

+ However, might need more info as $-r_A = k C_{A,s}$. But we don't have an expression for $C_{A,s}$.

+ This is where we use the fact that the reaction is external diffusion limited.

↳ The amount of A that diffuses to the surface reacts.

$$W_{A,r} = -r_A''$$

$$k_c(C_{A,b} - C_{A,s}) = \underbrace{(k' S_a)}_{k''} C_{A,s} \rightarrow$$

Can get an expression for $C_{A,s}$ which can be substituted in $-r_A$ to give $\frac{dF_A}{dW} = -r_A$

$$C_{As} = \frac{k_c C_A}{k_r + k_c}$$

$$\therefore \boxed{-r_{As}'' = \frac{k_c k_r C_A}{k_r + k_c}} \quad ; \quad k_{eff} = \frac{k_c k_r}{k_r + k_c}$$

We can do limiting cases to analyse what the rate is dependent on.

1) When $k_r \gg k_c$, $-r_{As}'' = k_c C_A$. When $Sc \uparrow$, we can just approximate $Sh = 0.6\, Re^{1/2} Sc^{1/3}$ (for a sphere)

$$\therefore k_c = 0.6 \cdot \underbrace{\frac{D_{AB}^{2/3}}{\nu^{1/6}}} \cdot \underbrace{\left(\frac{U}{d_p}\right)^{1/2}}$$

These will affect k_c and therefore the reaction rate of A.

eg: @ low U, diffusion limiting (big BL)
@ hig U, fast MT (could be reaction limiting)
} For a fixed U, decreasing d_p will result in high k_c (fast MT)

Chapt. 15 contd

1) Weiz-Prater Criterion: $C_{wp} = \eta\phi_1^2 = \dfrac{-r'_{A,obs}\,\rho_c R^2}{D_e C_{As}}$

$C_{wp} \ll 1$ No diffusion limitations within pellet

$C_w \gg 1$ $\eta \Delta \phi$

2) Mear Criterion: (for neglecting external diff. & Heat Transfer)

If $\dfrac{-r'_A)_b R_n}{k_c C_{Ab}} < 0.15$ } No external diffusion limitations

If $\left| \dfrac{-\Delta H_{Rx}(-r'_A)(\rho_b RE)}{h T^2 R_g} \right| < 0.15$

(R_g: gas constant; E: Activation energy)

$\dfrac{d^2 C_A}{dx^2}$ $x^2 = l^2\xi^2$ $\int^2 C$ $d($x $\dfrac{d^2 T}{d\xi^2}$

$l^2 d\xi^2$

CBE 142: Midterm 2 Review

+ Instantaneous Selectivity: $S_{D/U} = \frac{r_D}{r_U}$ D = desired, U = undesired

+ Overall Selectivity (Exit flow rates): $\tilde{S}_{D/U} = \frac{F_D}{F_U}$ D = desired, U = undesired

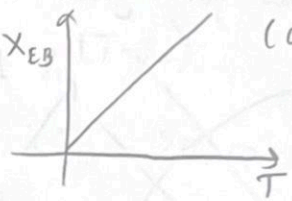
Chapter 11 [Adiabatic, Non-isothermal reactors]

1) $\Delta H_{Rx}(T) = \Delta H^\circ_{Rx}(T_R) + \Delta C_p(T - T_R)$

$\Delta C_p = c(C_p)_{products} - b(C_p)_{reactant}$ (stoichiometric ratio's)

2) $\dot{Q} - F_{A0}\sum \theta_i C_{p_i}(T - T_{i0}) - \left[\Delta H^\circ_R(T_R) + \Delta C_p(T - T_R)\right] F_{A0} X = 0$ [OP OG EB w/ $\dot{W}_s = 0$]

3) $X_{EB} = \dfrac{\sum \theta_i C_{p_i}(T - T_0)}{-\left[\Delta H^\circ_R(T_R) + \Delta C_p(T - T_R)\right]}$; $\dot{Q} = 0$; $\dot{W}_s = 0$



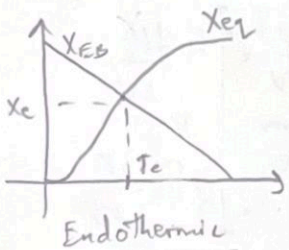
(works for all reactors)

$\Updownarrow$

5) $T = \dfrac{X\left[-\Delta H^\circ_{Rx}(T_R)\right] + \sum \theta_i C_p T_0 + X \Delta C_p T_R}{\sum \theta_i C_{p_i} + X \Delta C_p}$ $\xrightarrow[\text{Feed is pure A}]{\Delta C_p = 0}$ $T = T_0 + \dfrac{X\left[-\Delta H^\circ_{Rx}(T_R)\right]}{C_{P_A}}$

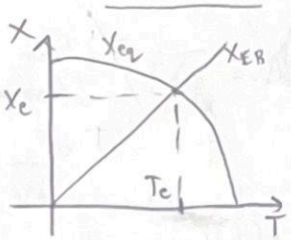
6) $K_c = K_c(T_2)\exp\left[\dfrac{\Delta H^\circ_{Rx}}{R}\left(\dfrac{1}{T_2} - \dfrac{1}{T}\right)\right]$

7) Equilibrium conversions [Express Conc. w.r.t X and use $K_c = \dfrac{[\text{products}]}{[\text{reactants}]}$]



Endothermic

Exothermic



$T_e \rightarrow$ adiabatic eq. Temp.

$X_e \rightarrow$ Adiabatic eq. conversion (Max conversion one can achieve for a given entering temperature T_0 & feed composition)

$X \not\propto$ flow rate ($X = f(K_c)$, $K_c = f(\Delta H^\circ, T)$)

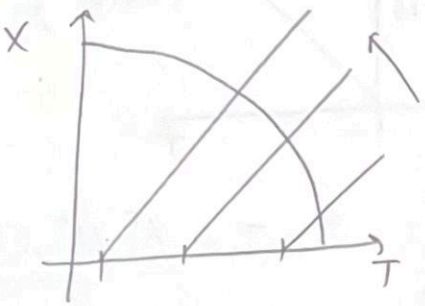
So X_e can only be affected by changing T (which can be done by adding inert or by reactor staging (Adding interstage cooling or heating)

8) Heat load: heat-exchange duty to cool the reacting mixture from T_0 to T

$$\dot{Q} = F_{A0} C_p (T_0 - T)$$

9) Eq. and reaction rate always go opposite: higher Feed T will have higher reaction rate ($r \propto k$; $k = f(T)$). However, $X_{EB} = \alpha (T - T_0)$ and the intersection of X_{eq} & X_{EB} will be lower giving lower X_e.

+ For a PFR & PBR, the X_e and X_{EB} changes along the reactor as the temp. in reactor also changes.
+ Although it may seem that a lower T will have higher X_e, the actual X of the reactor might even be 0 as the rate is very slow and the reactants fail to react for a given length / catalyst weight of reactor



Feed T is decreased

WHAT'S YOUR POINT? Questions
① Ex. 8.5 (Pg 304)

How are reaction rates calc for multiple gas phase?

Chapter 12 [Non-isothermal reactors]

For PFR: $\Delta \dot{Q} = U \Delta A (T_a - T)$

The di... energy ... $= 0$

$$\frac{dT}{dV} = \frac{\overbrace{r_A \Delta H_{Rx}}^{Q_g} - \overbrace{U_a (T - T_a)}^{Q_r}}{\Sigma F_i C_{p_i}} \quad (1)$$

product rule gives

$$H_i \frac{dF_i}{dV} + F_i \frac{dH_i}{dV}$$

from mole balance: $\frac{dF_i}{dV} = r_i$

$$C_p \frac{dT}{dV}$$

Exothermic: $Q_g > 0$

Endothermic: $Q_g < 0$

$$F_i = F_{A0} (\Theta_i + \nu_i X)$$

$$2) \quad \frac{dT}{dV} = \frac{r_A \Delta H_{Rx} - U_a (T - T_a)}{F_{A0} (\Sigma \Theta_i C_{p_i} + \Delta C_p X)} \quad (\text{as } f(X))$$

For PBR: $$3) \quad \frac{dT}{dW} = \frac{r'_A \Delta H_{Rx} - \frac{U_a (T - T_a)}{\rho_b}}{\Sigma F_i C_{p_i}}$$

$W = \rho_B \cdot V$, ρ_B = density of catalyst

Discover the best Pentel

Gaurav the legend : CBE 142

+ Michaelis Menton (All cases) → know how to derive it!

+ Energy balance: $\boxed{\frac{\partial E}{\partial t} = \dot{Q} + \dot{W}_s - F_{A0}\sum \theta_i \tilde{C}_{p_i}(T-T_0) - F_{A0}X_A[\Delta H^\circ_{rxn} + \Delta\hat{C}_p(T-T_r)]}$

Molar (ΔC_p)

$\tilde{C}_p = \frac{1}{T_2-T_1}\int_{T_1}^{T_2} C_p(T)\,dT$

↳ In our class is the same as C_{p_i}

$\Delta H^\circ_{rxn} = \Delta H_{rxn}$, $\Delta H_{rxn} \gg \Delta\tilde{C}_p(T-T_r)$

$\therefore X_A = \frac{\sum \theta_i \tilde{C}_{p_i}(T-T_0)}{\Delta H_{rxn}}$ } Adiabatic

$T \approx T_0 + \left(\frac{-\Delta H_{rxn}}{\sum \theta_i C_{p_i}}\right) X_A$ }

+ Wall-cooled CSTR:

$0 = UA(T_a - T) - X_A F_{A0}\Delta H_{rxn} - F_{A0}\sum \theta_i C_{p_i}(T-T_0)$

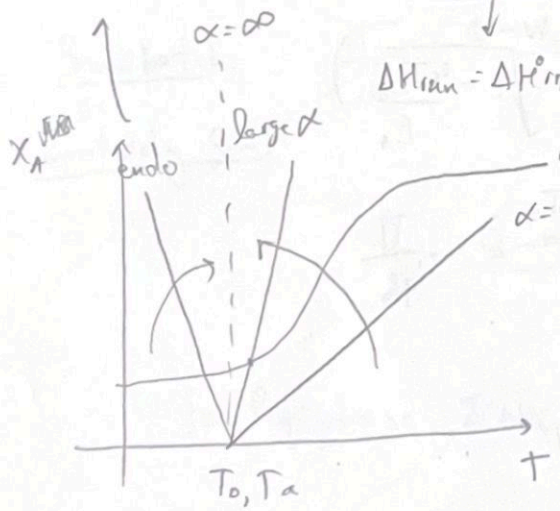
endo cannot have multiple S.S

$\therefore \boxed{X_A = \frac{\left(\frac{UA}{F_{A0}} + \overline{C_{p_0}}\right)(T^* - \bar{T})}{\Delta H_{rxn}}}$ $C_{p_0} = \overline{C_p} = \sum \theta_i C_{p_i}$

$T^* = \frac{T_0 + \alpha T_a}{1+\alpha}$

$\alpha = \frac{UA}{\overline{C_p} F_{A0}}$

$\Delta H_{rxn} = \Delta H^\circ_{rxn}$ $\Delta C_p \approx 0$



If $\alpha = 0$, $UA = 0$ ∴ adiabatic

All the α move towards $\alpha = \infty$ (for both endo & exothermic)

★ PFR: $q - F_{A0}\frac{d\bar{T}}{dV}\left(\overline{C_{p_0}} + \Delta\tilde{C_p}X_A\right) - F_{A0}\frac{dX_A}{dV}\left(\Delta H^\circ_{rxn} + \Delta\hat{C}_p(T-T_r)\right) = 0$

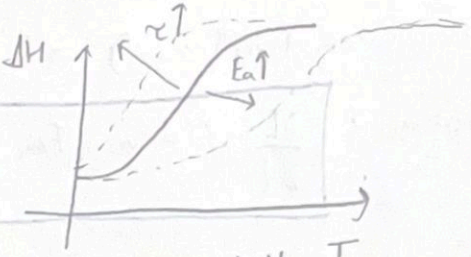
$q = U_{av}(T_a - T) = U\left(\frac{4}{D}\right)(T_a - T)$

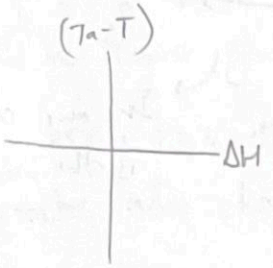
Only CSTR & exothermic reactions have multiple steady states.

$$R(T) = \bar{C}_p(1+\alpha)(T - T^*)$$

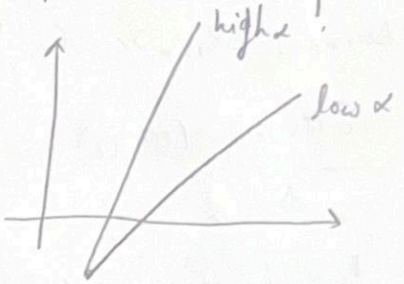
$$G(T) = \frac{-\Delta H^\circ_{rxn}\, k\tau}{1 + k\tau}$$

$$k = A_0 \exp\left(-E_a/RT\right)$$





You can still have R(T) for an endothermic reaction



Ergun eq:

$$\frac{dP}{dz} = -\frac{G}{g_c D_p}\left(\frac{1-\phi}{\phi}\right)\left(\underbrace{\frac{150(1-\phi)}{D_p}}_{\text{Laminar}}\mu + \underbrace{1.75G}_{\text{Turbulent}}\right)$$

$\phi \rightarrow$ void fraction

$G = \rho u = \frac{\dot{m}}{A_c}$

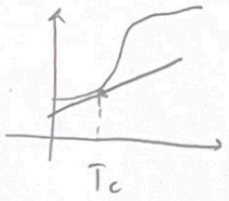
$u = \frac{Q}{A_c}$

$\downarrow D_p$ higher pressure drop

$\downarrow \phi \rightarrow$ lower pressure drop.

@ T^*, $R(T^*) = G(T^*)$

$$\left.\frac{\partial R}{\partial T}\right|_{T^*} = \left.\frac{\partial G}{\partial T}\right|_{T^*}$$



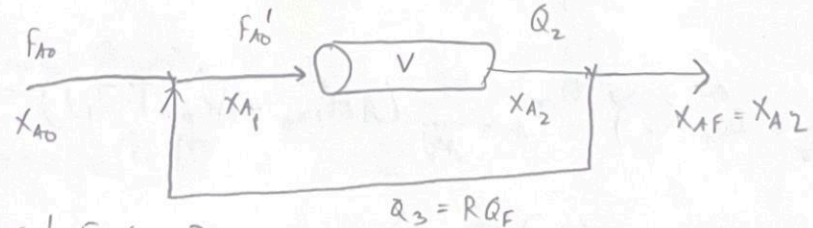
$$T^* = \frac{E_a}{2R}\left(1 - \sqrt{1 - \frac{4T_c R}{E_A}}\right)$$

$$T_c = \frac{\alpha T_a + T_0}{1+\alpha}$$

Shell balances on PFR:

+ Counter current
+ Auto thermal PFR

} Set up balance & BC's



Given, $R = \frac{Q_3}{Q_F}$

$$F_{A0}' = F_{A0}(R+1)$$

$$V = F_{A0}(R+1)\int_{\left(\frac{R}{R+1}\right)X_{Af}}^{X_{Af}} \frac{dX_A}{-r_A}$$

$$S_{D/U} = \frac{\text{rate D formed}}{\text{rate U formed}} = \frac{r_D}{r_U} = \frac{dC_D}{dC_U} \quad ; \quad \tilde{Y}_{D/U} = \frac{F_D}{F_{A0} - F_A} \quad ; \quad Y_{D/A} = \frac{r_D}{-r_A}$$

(Overall yield) (Instantaneous yield)

$$C_D = Y_{D/A}(C_{A0} - C_A)$$

$$C_D = -\int_{C_{A0}}^{C_A} Y_{D/A}\, dC_A$$

For CSTR: $\tilde{Y} = Y$

PFR: $\tilde{Y} = \frac{-1}{C_{A0} - C_A} \int_{C_{A0}}^{C_A} Y\, dC_A$

$A \xrightarrow{k_1} B \xrightarrow{k_2} C$ (B is desired)

BSTR: $t_{max\,B} = \frac{1}{k_2 - k_1} \ln\left(\frac{k_2}{k_1}\right)$

CSTR: $\tau_{max,B} = \frac{1}{\sqrt{k_1 k_2}}$

$\left(\frac{dC_B}{d\tau} = 0\right)$

Chapter 13

Semibatch or unsteady CSTR:

$$\frac{dT}{dt} = \frac{\dot{Q} - \dot{W}_s - \Sigma F_{i0}(H - H_{i0}) + (-\Delta H_{Rx})(-r_A V)}{\Sigma N_i C_{P_i}} \quad \text{(phase change)}$$

$$\frac{dT}{dt} = \frac{\dot{Q} - \dot{W}_s - \Sigma F_{i0} C_{P_i}(T - T_{i0}) + \left[-\Delta H_{Rx}(T)(-r_A V)\right]}{\Sigma N_i C_{P_i}} \quad \text{(No phase change)}$$

For liq phase: $\Sigma N_i C_{P_i} = N_{A0} \Sigma \Theta_i C_{P_i}$

Batch: $\frac{dT}{dt} = \frac{\dot{Q}_g - \dot{Q}_r}{\Sigma N_i C_{P_i}}$; $\frac{dT}{dt} = \frac{\dot{Q}_g - Q_r}{N_{A0}(\Sigma \Theta_i C_{P_i} + \Delta C_p X)}$ as $f(X)$

$$\dot{Q}_g = (-\Delta H_{Rx})(-r_A V)$$

$$Q_r = \dot{m} C_{P_c}(T - T_{a_1})\left[1 - \exp\left[\frac{-UA}{\dot{m} C_{P_c}}\right]\right]$$

4) Energy Balance on Exchangers:

(C = coolant)

Co-current Coolant

$$\frac{dT_a}{dV} = \frac{U_a(T-T_a)}{\dot{m}_c C_{P_c}}$$

Counter Current

$$\frac{dT_a}{dV} = \frac{U_a(T_a-T)}{\dot{m}_c C_{P_c}}$$

This is information on how the T_a is changing with the length of the reactor. (T_a is assumed to be no longer constant).

3) For a CSTR: (W by stirrer is neglected; we assume That there is no accumulation of energy in heat exchanger)

$$X = \frac{\frac{UA}{F_{A0}}(T-T_R) + \sum \theta_i C_{P_i}(T-T_0)}{[-\Delta H^\circ_{Rx}(T_R)]}$$

where $T_a \rightarrow$ coolant temp. (inlet T = outlet T)

$T_R \rightarrow$ reference T

$T_0 \rightarrow$ Feed inlet temp.

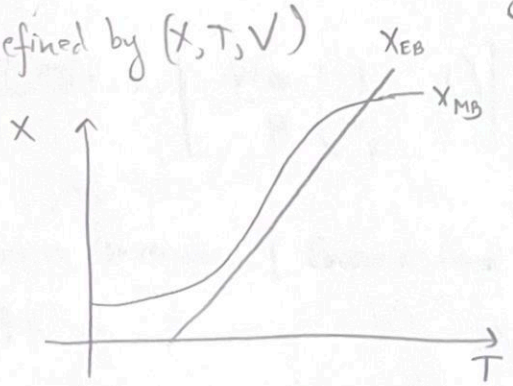
$$T = T_C + \frac{(-\Delta H^\circ_{Rx})X}{C_{P_0}(1+K)} \quad ; \quad K = \frac{UA}{(F_{A0}C_{P_0})} \quad ; \quad T_C = \frac{KT_a + T_0}{1+K} \quad ; \quad C_{P_0} = \sum \theta_i C_{P_i}$$

$$V = \frac{F_{A0}X}{-r_A(X,T)}$$

(Energy Bal) (Mole Bal.)

@ Steady state, No accumulation and EB = MB

CSTR is defined by (X, T, V)



$$X_{MB} = \frac{\tau k}{1+\tau k} \quad , \quad k = f(T)$$

$$X_{EB} = \frac{\sum \theta_i C_{P_i}(T-T_{i0})}{-[\Delta H^\circ_{Rx}(T_R) + \Delta C_p(T-T_R)]}$$

(Adiabatic)

CBE 142:

$C_j = \frac{N_j}{V}$ $\quad$ $C_j = \frac{F_j}{\bar{V}}$

BSTR

① $\boxed{r_j \int V = \frac{dN_j}{dt}}$

$V = \text{const}: \quad \frac{dC_j}{dt} = r_j$

$V \neq \text{const}: \quad r_j = \frac{1}{V}\frac{dN_j}{dt}$

$\Rightarrow r_j = \frac{dC_j}{dt} + \frac{C_j}{V}\frac{dV}{dt}$

Design eqs:

* $r_A = -C_A \frac{dX}{dt}$
* $\frac{t}{C_{A0}} = \int_0^{X_A} \frac{dX_A}{(-r_A)}$
* $dt = \frac{-N_{A0}\,dX}{V \cdot r_A}$

Same

PFR

① $\boxed{r_j = \frac{F_j - F_{j0}}{\Delta V}}$

$r_j = \frac{dF_j}{dV}$

$\bar{v}_0 = \text{const}: \quad r_j = \bar{v}_0 \frac{dC_j}{dV}$

as $\frac{dV}{\bar{v}_0} = d\tau$, $d\tau = \frac{dC_j}{r_j}$

$\bar{v} \neq \bar{v}_0$, $dV = d\tau\, \bar{v}(x,y)$

Design eqs:

* $\frac{V}{F_{A0}} = \frac{\tau}{C_{A0}} = \int_0^{X_A} \frac{dX_A}{-r_A}$
* $r_A = -F_{A0}\frac{dX}{dV}$
* $V = \int_0^X \frac{F_{A0}\,dX_A}{-r_A}$

CSTR

① $\boxed{r_j \int V = F_j - F_{j0}}$

$\bar{v}_0 = \text{const}: \quad V = \frac{v_0 (C_{j0} - C_j)}{-r_j}$

$(V = \tau \bar{v}_0)$

$\tau = \frac{V}{\bar{v}_0}$

Design eqs (relating X)

* $\tau = \frac{C_{A0} X_A}{-r_A}$
* $V = \frac{F_{A0} X_A}{-r_A}$

Expressing C as $f(X)$: $\quad C = \frac{N_j}{V} = \frac{F_j}{\bar{V}}$

→ Batch: $C_A = \frac{N_{A0}(1-X)}{V}$ If $V = V_0$ $\therefore C_A = C_A(1-X)$

→ Flow: $C_A = \frac{F_A}{\bar{V}} = \frac{F_{A0}(1-X)}{\bar{V}}$;

Gas

$\bar{V} = \bar{V}_0 (1+\varepsilon X)\frac{P_0}{P}\frac{T}{T_0}$

$C_A = C_{A0}\frac{(1-X)}{(1+\varepsilon X)}\frac{P}{P_0}\frac{T_0}{T}$

Liquid

$\bar{V} = \bar{V}_0$

$C_A = C_{A0}(1-X)$

$D = \tau \cdot k$

↑

Damkohler no.

Slit to put

+ Rate two times! + Enzyme.

+ Rate diagram

$V = V_0 \frac{F_T}{F_{T0}}$

$r_a = -k_1[A][B] + k_{-1}[I_1] + 2k_2[A]^2[I_1]$

$= -k_1[A][B] + (k_1 + 2k_2[A]^2)[I_1]$

Stoichiometry For a given reaction $aA + bB \rightarrow cC + dD$ we can construct a stoich table (having initial moles, change in moles after reaction & remaining no. of moles)

If A is limiting reagent, then $N_C = N_{C_0} + \frac{c}{a}(N_{A_0} \cdot X)$ ← conversion

$$\delta = \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1\right) = \frac{\text{Change in the total no. of moles}}{\text{Moles of A reacted}}$$

$$\therefore N_T = N_{T_0} + \delta(N_{A_0} X)$$

To get conc.: $C_C = \frac{N_C}{V} = \frac{N_{C_0} + (c/a) N_{A_0} X}{V}$

If $V = V_0$

$$= \frac{N_{A_0}\left(N_{C_0}/N_{A_0} + (c/a)X\right)}{V_0}$$

$\Theta = \frac{N_{C_0}}{N_{A_0}}$

$$\boxed{C_C = C_{A_0}\left(\Theta_C + \frac{c}{a}X\right)}$$

$\Theta_C = 1$ (reaction is equimolar)

$\Theta_C = \frac{b}{a}$ (stoichiometric)

$$\boxed{\Theta_i = \frac{N_{i_0}}{N_{A_0}} = \frac{C_{i_0}}{C_{A_0}} = \frac{y_{i_0}}{y_{A_0}}}$$

Therefore for a constant volume reactor we can express concentration as a function of conversion

$\Downarrow$

$\therefore -r_A = f(X_A)$

Don't forget to chose the limiting reagent as the basis for calculation (A)

From the stoich table, we get total no. of moles as:

$$F_T = F_{T_0} + F_{A_0}\delta X$$

$$\div F_{T_0} \Rightarrow \frac{F_T}{F_{T_0}} = 1 + \underbrace{\frac{F_{A_0}}{F_{T_0}}}_{y}\delta X = 1 + \underbrace{y_{A_0}\delta}_{\varepsilon} X$$

just the stoich balance which gives moles of species

$$\delta = \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1\right)$$

$$\boxed{\varepsilon = y_{A_0}\delta}$$

$\varepsilon\varepsilon = \frac{\Delta \text{Total moles for complete conversion}}{\text{Total moles fed}}$

...w systems are very similar to Batch system.

+ Varying V and ν_0

The same stoich tables work! (Only have to take care when considering concentration - as a $f(X_A)$)

→ Most gas-phase reactions: (j = species; T = total)

$$C_T = \frac{P}{ZRT} \quad \left(C_{T0} = \frac{P_0}{Z_0 R T_0} \right) \quad \therefore \quad \frac{V}{V_0} = \frac{(F_T/C_T)}{(F_{T0}/C_{T0})}$$

$$C_j = \frac{F_j}{\nu} = \frac{F_T}{\alpha} = \underbrace{\left(\frac{F_{T0}}{\nu_0}\right)}_{C_{T0}} \left(\frac{F_A}{F_T}\right)\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right)$$

$$\boxed{C_j = (C_{T0})\left(\frac{F_j}{F_T}\right)\left(\frac{P}{P_0}\right)\left(\frac{T_0}{T}\right)}$$

$$\boxed{\frac{V}{V_0} = \frac{F_T}{F_{T0}}\left(\frac{T}{T_0}\right)\left(\frac{P_0}{P}\right)}$$

→ $\nu = \alpha$

→ $C_j = C_{T0}\, y_j\, P \frac{T_0}{T}$

As defined earlier, $\frac{F_T}{F_{T0}} = (1 + \varepsilon X)$ & $F_j = F_{A0}(\theta_j + \gamma_j X)$ ← general eq from stoich table.

2: Things to know

: dimensionless number that can give a quick estimate of the degree of conversion that can be achieved CSTR.

$$Da = \frac{-r_{A0}V}{F_{A0}} = \frac{\text{Rate of reaction at Entrance}}{\text{Entering flow rate of A}} = \frac{\text{"Reaction rate of A"}}{\text{"Convection rate of A"}}$$

For first order irreversible reaction:

$$Da = \frac{-r_{A0}V}{F_{A0}} = \frac{k_1 C_{A0} V}{-v_0 C_{A0}} = \tau k_1$$

For 2nd orde :

$$Da = \frac{-r_{A0}V}{F_{A0}} = \frac{k_2 C_{A0}^2 V}{v_0 C_{A0}} = \tau k_2 C_{A0}$$

For irreversible reactions:

$Da < 0.1 \Rightarrow X < 0.1$

$Da > 10 \Rightarrow X > 0.9$

CSTR in series

In - out + generation = Accumulation

$$F_{A0} - F_A + r_A V = 0 \Rightarrow r_A V = F_{A0} - F_{A0}(1-X) \qquad r_A = -kC_A$$

$$r_A V = F_{A0} X$$

$$V = \frac{F_{A0} X}{r_A} \quad (\div v_0) \Rightarrow \tau = \frac{C_{A0} v_0 X}{v_0 r_A}$$

Conversion: $F_A = F_{A0}(1-X)$

$C_A = C_{A0}(1-X)$

$$\tau = \frac{C_A X}{(1-X) r_A} = \frac{X}{k(1-X)}$$

$$\boxed{\tau = \frac{X}{k(1-X)}}$$

$$\tau k = \frac{X}{(1-X)}$$

$$\boxed{X = \frac{\tau k}{1+\tau k}}$$

Now,

$$C_A = C_{A0}\left(1 - \frac{\tau k}{1+\tau k}\right)$$

Final Crib Sheet

Batch

$$\frac{dT}{dt} = \frac{(-\Delta H_{Rx})(-r_A V)}{\sum N_i C_{P_i}}$$

[Adiabatic, $W_s = 0$]

for liq phase: $\sum N_i C_{P_i} = N_{A0} \sum \Theta_i C_{P_i}$

+ Unsteady
+ $W_s = 0$

$$\frac{dT}{dt} = \frac{\overbrace{(-\Delta H_{Rx})(-r_A V)}^{\dot{Q}_g} - \overbrace{\dot{m} C_{P_c}(T - T_{a_1})\left[1 - \exp\left(\frac{-UA}{\dot{m} C_{P_c}}\right)\right]}^{\dot{Q}_r}}{\sum N_i C_{P_i}}$$

(In terms of $f(X)$)

$$\frac{dT}{dt} = \frac{\dot{Q}_g - \dot{Q}_r}{N_{A0}\left(\sum \Theta_i C_{P_i} + \Delta C_p X\right)} \; ; \; N_{A0}\frac{dX}{dt} = -r_A V$$

Semi-Batch

$$\frac{dT}{dt} = \frac{\dot{Q} - \dot{W}_s - \sum F_{i0}(H - H_{i0}) + (-\Delta H_{Rx})(-r_A V)}{\sum N_i C_{P_i}} \quad \text{(phase change)}$$

$$\frac{dT}{dt} = \frac{\dot{Q} - \dot{W}_s - \sum F_{i0} C_{P_i}(T - T_{i0}) + \left[(-\Delta H_{R_2}(T))(-r_A V)\right]}{\sum N_i C_{P_i}} \quad \text{(No phase change)}$$

$(\dot{W}_s = 0)$ →

$$\frac{dT}{dt} = \frac{\overbrace{(r_A V)(\Delta H_{Rx})}^{Q_g} - \overbrace{\sum F_{i0} C_{P_i}(T - T_{i0}) + \dot{m} C_p [T - T_{a_1}]\left[1 - \exp\left(\frac{-UA}{\dot{m} C_{P_i}}\right)\right]}^{Q_r}}{\sum N_i C_{P_i}}$$

For multiple reactions: $Q_g = V \sum_{i=1}^{q} r_{ij} \Delta H_{Rx\,ij}$; $Q_r = UA(T - T_a) + \sum F_{j0} C_{P_i}(T - T_0)$ (High coolant flow rates)

CSTR W is neglected

$$V = \frac{F_{A0} X}{-r_A(X,T)} \; ; \; X^{CB} = \frac{\frac{UA}{F_{A0}}(T - T_a) + \sum \Theta_i C_{P_i}(T - T_0)}{\left[-\Delta H^{\circ}_{Rx}(T_R)\right]} \; ; \; X^{EB} = \frac{\left(\frac{UA}{F_{A0}} + \overline{C_{P_0}}\right)(T^* - T)}{\left[\Delta H^{\circ}_{rxn} + \Delta \tilde{C}_p (T - T_A)\right]}$$

$$\alpha = UA / \overline{C_P} F_{A0}$$

$$T^* = \frac{T_0 + \alpha T_a}{1 + \alpha}$$

$$\overline{C_{P_0}} = \sum \Theta_i C_{P_i}$$

$$\underbrace{-F_{A0} \Delta H^{\circ}_{rxn} X_A^{MB}}_{G(T)} = \underbrace{F_{A0} \overline{C_{P_0}}(1+\alpha)(T - T^*)}_{R(T)}$$

$G(T) = \Delta H^{\circ}_{rxn} X_A^{MB}$

Do we have F_{A0}??

Is this correct?

$$Q_g = \Delta H_{rxn}(-r_A V)$$

$\overline{F_{T0}}$